

2. General Settings for Full Closed Loop

V.4.2

Please note that with autoISF you are in an early-dev. environment, where the user interface is **not optimized for safety** of users who stray away from intended ways to use. Good safety features exist, but these are only as good as the development-oriented user understands and implements them. This is not a medical product, refer to disclaimer in [section 0](#)



2.1 SMB Range Extension
2.2 Max and Min autoISF Ratio
2.3 SMB Delivery Ratio
2.4 iobTH
2.5 Eating Soon TT ?
2.6 Other settings in AAPS for autoISF FCL

[Available related case studies:](#)

Case study 2.1: (nothing available yet)

When in Hybrid Closed Loop, big boli were given by the user, and the loop had no business to give big ones on top. But this is fundamentally different now.

So, first we must **enable our loop to overcome the narrow safety restrictions for SMB sizes** that were appropriate in HCL.

Evidently, this could become dangerous. Please set your iobTH ([section 2.4](#)), and observe all suggestions made in this FCL-e-book *) , and in the github pages of the developers.

See also: https://github.com/ga-zelle/autoISF/blob/A3.2.0.2_ai3.0/How-to-get-larger-SMBs.pdf
/or newer branch/

***) Caution:** This entire e-book is about Full Closed Looping. **In case you intend to work with giving boli**, many suggestions made - notably in this section 2 (and in section 4) – should **not** be followed. You would have to **do extra research**, on your own data, how your bolus changes things. Mainly use the documentations referred to in [section 3](#), then. See also remarks in [section 4.1](#) and [section 7](#)

2.1 SMB Range Extention

(preferences/OpenAPS SMB/autoISF settings/smb delivery settings/smb_max_range_extension)

2.1.1 Standard scenario: SMB sizes based on 5 minute loop calculations

Full Closed Looping requires bigger SMB sizes. Setting **120 minutes** of basal as max. SMB size as enabled in AAPS Master is necessary, but will rarely suffice.

In AAPS Preferences/OpenAPS SMB/autoISF settings/smb delivery settings, set SMB/UAM max range extension (**smb_max_range_extension**) to **2.0**. That doubles the allowed max. size, to four hours of your profile basal, *for a start*.

Even better, you could determine an estimate for *your* initial setting as in the following description, I will use the symbol, $\dots$, to denote where you would use *your* numbers. My numbers that I use for the same situation will be in parentheses (U).
In full closed loop, once the bg starts rising, you want to get at least half of your required meal bolus within 10 minutes, through 2 SMBs. To do that, you need $\dots$ U (2 U) per SMB on average, and because the bolus sizes tend not to be equal when requested by the Loop, you should have at least $\dots$ U (3 U) as the allowable SMB size. Your hourly basal is around $\dots$ U (0.6 U), i.e. AAPS Master will allow a max. 2 times that hourly basal which = $\dots$ U (1.2 U) per SMB. To reach the intended $\dots$ U (3 U) therefore you should set your smb_max_range_extention to $\dots$ ($2.5 = 3 \text{ U} / 1.2 \text{ U}$)

The profile helper in [section 4.8](#) might be available for doing this calculation, and for a cross-check.

If you have a very low hourly basal rate, extensions bigger than 3.0 can result (maximum you can set is 5).

Note: Elevated insulin needs in phases of elevated insulin resistance probably will be managed with **>100% profile adjustments**. Then, profile basal gets elevated accordingly, and thus **will automatically allow increased SMB size**.

Watch out (in your AUTO ISF tab, or using the emulator, [section 10](#)) whether you often run into a limitation by your set smb_max_range_extention. For instance, your **attempts to increase initial SMB** sizes via elevated smb_delivery_ratio ([section 2.3](#)) and elevated bgAccel_ISF_weight ([section 4.2](#)) **might get cut by a too low smb_max_range_extention**.

The SMBs your loop requests could get reduced in size also by other safety settings, notably by your autoISF_max setting (see [section 2.2](#))

2.1.2 Special Libre 3 (1 minute) scenario with up to 5 SMBs per 5 minutes

When receiving bg values every minute, and adjusting insulin delivery accordingly in smaller steps, probably 120 minutes of basal per SMB suffices.

Watch whether your system actually can process 5 loop calculations (and potentially give 5 SMBs) in 5 minutes. Then judge (similar like shown in [section 2.1.1](#)), what maximum SMB size you would like to see.

75 Probably you can leave SMB_range_extension at default **1.0** untouched (but select higher if you have an
76 extremely low profile basal in some of your meal hours, or if your loop “misses” many of the opportunities
77 to make an adjustment, every minute).

78 Integration of 1-minute values is new from autoISF 3.0.1 (May 2024). Please watch your AUTO ISF tab
79 (difficult in 1 minute segments => make screenshots, or employ Emulator!), and report experiences / stay in
80 touch with other users.

81

82 2.2 Max and Min autoISF Ratio

83 (preferences/OpenAPS SMB/autoISF settings/autoISF_min and autoISF_max)

84

85 For a start, set **autoISF_max** = 2.0 . –

86 Your CGM (1 or 5 minute re-calculations) should not matter for this parameter.

87 This allows *up to doubling* of ISF aggressiveness if “requested by the “... _ISF_weights” (see
88 [section 4](#)). This is just a first step.

89 You may have to elevate autoISF_max further, later, if your attempts to tune the ...ISF_weights
90 (section 4.) often run into a limitation by your set autoISF_max

91

92 For situations of increased insulin sensitivity (less insulin need), you must specify, in your settings
93 (preferences) also what your *lower* limit (for weakening of ISF, compared to profile_ISF) may be.

94

95 **autoISF_min** should be set to 0.3. Again, see whether you ever run into that limit, that your loop
96 e.g. “would like” to act softer, but bounces against that set limit.

97

98 Do not keep autoISF_min at 0.5 or even higher, because that would for instance preclude
99 later, that your exercise setting can strongly “soften” your loop.

100

101 2.3 SMB Delivery Ratio

102 (preferences/OpenAPS SMB/autoISF settings/smb delivery settings/smb_delivery_ratio)

103

104 2.3.1 Standard scenario, using 5 minute loop calculations

105

106 Use the *fixed* **smb_delivery_ratio** and increase the setting (from AAPS default 0.5) to 0.6 or 0.7.

107

108 In AAPS Preferences, the smb_delivery_ratio *can* also be set *dynamically* (changing with
109 glucose level). - This seems a feature geared more towards hybrid closed loop applications.
110 (In FCL, we like strong loop aggressiveness at low (but rising) bg, not “wait for” high bg).

111 So, set your ratio to 0.6 or 0.7 before doing any _weights tuning. Your choice will magnify every
112 SMB, also in phases where you actually want less (e.g. with a “jumpy” CGM!), so do not
113 exaggerate.

114

115 0.6 gets you 20% , 0.7 gets you 40% more insulin 5 minutes earlier, which is a good thing in FCL
116 where you are late with your first meal insulin. But you do not get 20-40 % more really: You
117 gradually will receive the full insulinReq, - only in increased % increments, and some of it 5 -10
118 minutes earlier.

119 The delivery ratio is per se not changing the insulinReq, it just defines what portion gets delivered
120 now vs 5 or 10 minutes later... if the BG trend keeps up...

121 In that way, keeping the number closer to 0.5 protects against a jittery CGM mostly.

122 As in FCL we have an above-avg CGM quality, we can safely go for 0.6 or 0.7 SMB delivery ratio as
123 kind of our tuning baseline.

124 In case you use very strong **smoothing** (e.g. exponential smoothing of incoming CGM values by
125 AAPS) you probably can afford to go more towards 100%. This would make up for probably only
126 some of the “time lost by smoothing” , regarding getting iob up at first signs of bg rising.

127 But, the closer you set towards 100%, the more have an eye on how each single bg value AAPS
128 works with triggers SMBs.

129 **The jumpier your CGM, the closer remain near 0.5!**

130

131 **Note that it is generally a flawed strategy, to boost “*across the board*” with things like**
132 **(*always*) a 100% delivery rate, and then limit problems from it via reducing the *dynamic***
133 ***range of other highly important parameters* (for example, of autoISFmax). While, taken**
134 **together, these two elements of your flawed strategy would keep you safe, they make your FCL**
135 **less dynamic, which essentially will translate into giving up a couple of %TIR.**

136

137 In case you had tuned with a 100% SMB delivery ratio, and now go lower:

138 I would not expect major re- tuning required, but look into implications for the set iobTH%:

139 **A 100% SMB delivery ratio** made you often **bounce over iobTH in a more "nervous" loop**. You
140 now could fine tune that more sensibly, probably elevate the iobTH even (and maybe also , slightly,
141 the accel weight...which, for safety against your wild 100% setting, had maybe to be lowered
142 before)(Or, If you were unsafe before, leave iobTH where it was, and you are safer now, with a
143 lower SMB delivery ratio)

144

145

146

147

148 2.3.2 Special Libre 2 or 3 scenario, when using 1 minute values

149

150 The recommendation there is to set the data flow Libre -> Juggluco -> AAPS -> exponential
151 smoothing, and use a **SMB delivery ratio under 50%..**

152

153 To understand how the 1-minute values are used

154 • for the (still) 5 minute incremental loop calculations, now done every minute, looking back
155 how the last 5 minutes (evtl. interpolated) went

156 • for the parabola fit based acceleration detection

157 please consult the related section (last chapter) in the developers' Quick Guide:

158 https://github.com/ga-zelle/autoISF/blob/A3.2.0.4_ai3.0.1/autoISF3.0.1_Quick_Guide.pdf

159

160 Also, in this very new application, it is especially important to share experience with other FSL
161 users in Discord -> <https://discord.gg/tamvhh57Xs>

162

163

164 2.4 Safety Against too Aggressive Settings: iobTH%

165 (preferences/OpenAPS SMB/ autoISF settings/smb delivery settings/iob_threshold_percent

166 ...which gets multiplied with preferences/OpenAPS SMB/Maximum total IOB OpenAPS can't go over (U)

167

168 A safety net is needed because autoISF shoots big SMBs when glucose levels begin to rise; but
169 you do not want to bounce into your ultimate maximum total iob (iobMAX) safety setting too often.

170

171 Similarly, this safety net is needed also if using autoISF in Hybrid Closed Loop (HCL), where, after a
172 user bolus already provided some iob, autoISF could add too big SMBs to be safe.

173

174 Therefore, we install an **iob threshold** (iobTH) which, **when**, and as long as, **exceeded**, **shuts**
175 **SMBs off.**

176

177 **Step 1: In Preferences, set the SMB toggle for even /odd targets to "ON"**

178

179 autoISF 3.0.1 demands (as we recommend for FCL also for other reasons, see e.g. later in [section 5.1.2](#)
180 and [5.1.3](#)) that concurrently, in AAPS / Preferences / **Open APS SMB / autoISF settings / SMB delivery**
181 **settings:**

182

183 Enable alternative activation of SMB depending on current target *) **ON**

184 *) *previous autoISF versions* allowed different settings, now it is same setting,
185 for profile target and for TT

186 Step 2: In Preferences, set your default iobTH_percent

187

188 iobTH is a parameter in AAPS preferences, defined there as fraction of your set maxIOB:

189 /OpenAPS_SMB/autolSF_settings/Full_Loop_settings:

190 Percentage of maxIOB above which SMBs are disabled (iob_threshold_percent,)

191

192 Step 2.1: Solidify your maxIOB

193

194 First, check whether your **maxIOB** is set reasonably in AAPS Preferences / OpenAPS SMB /

195 Maximum total IOB OpenAPS can't go over (U).

- 196 • Input a figure (units) slightly above the max level of iob you had ever needed in your past looping
197 history (also considering times of elevated insulin resistance you occasionally may have had to deal
198 with); set maxIOB to that value.
- 199 • If you do not have “old” data at hand: **Max iob** would be the sum of hourly profile basal, plus the max
200 meal bolus you might need (g carbs digested in first ~ 2.5 h divided by IC), plus correction bolus for
201 elevated value (max seen BG at meal starts minus target BG divided by ISF). Then take the result
202 times a factor, e.g. times 1.2 in case you see on some days 20% higher insulin sensitivity, and will
203 use 120% profile (switch, or driven by Autosens max, or by other settings, see section 5 and 6,
204 later).

205

206 Step 2.2: Identify your max iob need in big meals

207

208 Now look at your meal spectrum, and what bolus size, and iob level (including from SMBs; in HCL
209 or FCL) was useful *) in high carb meals to control your glucose. *(For instance, the author needed
210 up to 8U early-on in big meals in HCL; and he has TDD near 40 U, and maxIOB set to 10 U).*

211 *) useful level = **iob needed** for the meal; iob may in time have gotten even higher. However, if, in
212 the end, to prevent a hypoglycemia, you had to consume 15 g carbs, then deduct 15 g / (your IC)
213 from that even higher iob **you actually did temp. have**. Example: $15 \text{ g} / (10 \text{ g/U}) = 1.5 \text{ U}$

214

215 Step 2.3: Set your iob_threshold_% in AAPS/Preferences

216

217 Assume you want to approach no more than about 75% of that iob level (that would be useful to
218 have at big high carb meals) via rapid SMB „fire“, after a meal related bg rise is seen *(then, for
219 instance, reduce from 8 U to 6 U).*

220

221 Then calculate your setting for **iob_threshold_percent** in AAPS / Preferences:

222 = desired total iob given via SMBs before bg peaks / iobMAX

223

224 Enter the according percentage in /Preferences

225 (In the example it would be $= 6 U / 10 U = 0.6$; which means to enter 60 as percentage in
226 /preferences).

227

228 The iobTH then is calculated as follows:

229
$$\mathbf{iobTH} = \mathbf{iobTH_percent} \times \mathbf{maxIOB}$$

230

231 Step 2.4: High-carbers may need to reduce their iob_threshold_percent a bit, to factor in that *the*
232 *last “allowed” SMB can shoot above iobTH:*

233

234 • The last SMB given **can exceed** that threshold by up to **+30%** of the effective iobTH.

235 This is desirable because it allows higher iob at big high carb meals (where SMB size, when
236 approaching iobTH, is still big); at lower carb meals either iobTH will not be reached anyways, or
237 SMBs are quite small when reaching iobTH and will not shoot over by much.

238 A big SMB that would shoot over by more than +30% will be cut at 130% iobTH.

239 • Until iob falls below effective iobTH, only %TBRs supply more insulin, if the loop calculates that
240 more iob is still required.

241 • In low carb meals, that iobTH level should not be reached => the autoISF parameters
242 („weights“) need to be tuned carefully, so SMB sizes are *not always* huge and bounce against
243 the iobTH restriction, but show different behavior for different meals

244 • Note that when operating with an *even elevated* bg target (>100 mg/dl), iobTH can only be
245 exceeded by **+20%** (“loop at medium power”). This makes sense, notably in an exercise
246 context (...in which *the iobTH per se* also gets automatically lowered, as later discussed in
247 [section 6.1.3](#)).

248

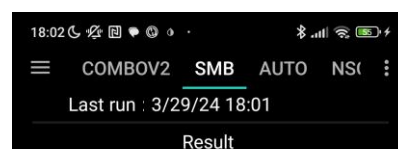
249 Step 2.5: At some later stage, come back to fine-adjust your percent setting in /Preferences,,
250 factoring in that it can be auto-modulated.

251

252 • autoISF 3.0 and higher contains a function to **auto-adjust** iobTH with TT set: Dynamic iobTH
253 ([section 6.1.3](#)). In your initial tuning, just set a iobTH_percent that is good-enough on your
254 average day. A **low** (e.g. EatingSoon) TT can automatically **elevate** iobTH.

255 • An **elevated** (e.g. exercise) TT can automatically **lower** iobTH, which is highly desirable **for**
256 **exercise**. The formula for the resulting effective iobTH is complicated, especially when the
257 exercise mode is also activated. See [sections 3.3](#) and [6.1.3](#), and example in [case study 6.2](#)

258

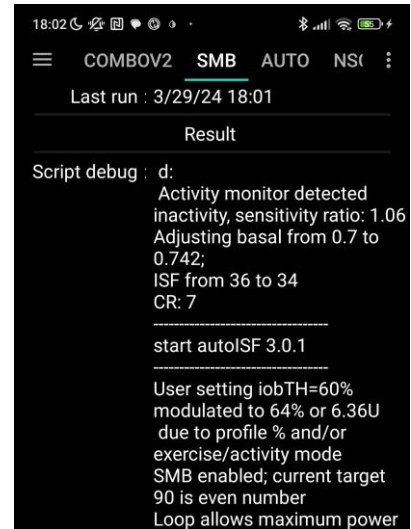


- The resulting **effective iobTH** can be seen in the AUTO ISF tab. From autoISF 3.0.1 onwards, the **AUTO ISF tab** starts with the Result section, and right below the / start autoISF headline, the resulting modulated iobTH is given:

The iobTH can also get temporarily modulated.

The example on the right shows it **elevated** in a case of detected **in-activity**:

In the exercise mode, it will automatically become radically *reduced* to limit iob during sports (see [section 6](#), “dynamic iobTH”).



- Likewise, a **%profile** setting < 100% (or > 100%) will proportionally lower (or elevate) the iobTH that will be used.

$$\text{effective iobTH} = \% \text{ temp.profile} \times \text{iobTH}$$

After the temporary % profile expired, it will automatically revert to your originally set iob_threshold_percent.

Off-topic note, regarding the **effective ISF (“sens”)**:

In the AUTO ISF tab, *above the “start autoISF.” line*, the profile ISF is given (“ISF unchanged”), eventually with adaptation by activity monitor (“adjusting ...ISF from ... to .. “ ?) or by a TT (“adjusting ...ISF from ... to ..”) or by a %temp. profile set (“unfortunately” still called “ISF unchanged” then).

Then follows the autoISF section explaining in detail how the recently encountered bg curve characteristics suggest adaptations, and what overall the conclusion is (“final ISF factor”, calculated following the flowcharts as explained in detail in section 03.).

Below the autoISF section, the effective ISF (sens) results from dividing the (unchanged or adapted) ISF *prior to* “start autoISF”, with the determined “final ISF factor” at the end of the autoISF section of the AUTO ISF tab.

Example given in [section 5.4.5](#)

If both, a % profile and a TT are set, both effects multiply.

We shall see later, how this opens nice avenues for exercise management, where we like to strongly limit how high iob shall be allowed to go. Example given in [case study 6.3](#).

All above discussed settings must be made in AAPS/preferences

- except for the **temporary modulations** of sensitivity, which can be done from the AAPS home screen via %profile or via TT inputs

(This will in detail be presented in [section 5](#) or, regarding exercise button, in [section 6.1.3](#)).

Regarding activity monitor see [section 6.6](#)).

Setting different iobTH via an Automation.

If you do have situations where you rather use a different iobTH_percent than set in your profile (or than resulting in an eventual automatic-modulation), you can change it also via an Automation.

Caution: This will overwrite your iob_threshold_percent until you restore it, manually (in /Preferences), or via another Automation (see e.g. [section 5.1.4](#)).

For this reason, preferably work with the temporary adaptations via %profile, TT and exercise mode as discussed above.

Next steps:

Before going first time into FCL mode, you must **first**

- check proper AAPS settings according to [section 2.6](#)
- **make additional settings in /preferences/Open APS SMB/autoISF**, when you get to [section 4](#).

Only after you also went through [section 4](#), you can **enter/exit FCL** (for initial tuning, or for everyday utilization) via

- in AAPS/Preferences/OpenAPS SMB/autoISF settings/"**Enable ISF adaptation by glucose behavior**" **ON** / **OFF**

In your multi-week FCL set up phase you will quickly notice that changing this setting back and forth "all the time", in your initial project weeks, is not convenient.

A much easier way to "**switch off**" **FCL aggressiveness** is to set an odd-numbered bg target (and an even target again, when you want normal FCL aggressiveness again).

With odd bg targets (if in mmol, odd decimal), you run in "AMA" mode, still with autoISF boosted %TBRs, but much milder acting because you will not receive any SMBs. See discussion in [section 5.1](#)

- An improved solution might become in the future available via a User Interface upgrade (described in [section 5.3.1](#)): Switching between **FCL and HCL** by just tapping on the **violet/ green closed loop icon** of your AAPS home screen - after this feature becomes integrated in a future autoISF version update.

330 2.5 EatingSoon TT ?

331

332 *FCL works in principle also without setting an EatingSoonTT. You could skip this section for now*
333 *(and revisit later in case you miss any performance via the totally hands-off way). Or you could*
334 *also just jump into the “preferred solution” presented on the next page...*

335

336 Your FCL works best if you start meals at below-target glucose values, and ideally have a bit of
337 positive iob at meal start. Also, a low temp. glucose target helps making SMBs (that „aim at it“) a bit
338 bigger.

339

340 Setting an EatingSoonTT well ahead of meal start therefore is *in principle* a good idea.

341

342 • *If you have relatively fixed meal time slots in the 24 hours of the day, you could actually set the*
343 **target glucose values in your profile** accordingly, e.g. a 11-15h profile target of 76 mg/dl if
344 you “almost always” start a lunch between 11:45 and 14:30h. *(Should you skip eating, nothing*
345 *bad would happen. Only if you do exercise instead, you likely will need a snack).*

346 • Note that you can opt for “turbo charging” your loop a bit with lowered ISF via making use of
347 features as discussed in [section 2.6](#) at points 5. and 7.

348 • *If you have rather irregular habits*, it might be worthwhile to either

349 ○ **manually** set an **EatingSoonTT** (which is quite time-uncritical) well before the start of a
350 meal, or even (latest) when the first SMB is about to be triggered by your loop – or

351 ○ do this via an **Automation** (see next page) - or

352 ○ just forget about it. Setting an EatingSoonTT would often have only a minor effect,
353 as: your meals are usually spaced by a couple of hours => glucose should not be
354 elevated as you approach the next meal. (Or, if glucose is still on its way down, it comes
355 with insulin on board and supports momentum for acceleration after the next meal
356 start. ...

357 Speaking about acceleration: In autoISF FCL, first SMBs driven by bgAccel_ISF are decisive.

358 They get cut in size, in case bg sits below target (formula see in [section 4.2.4](#)). Hence, a low

359 temporary (even-numbered) bg target secures earliest-possible ramping up of iob after meal starts.

360 There is a *fully automatic* way to integrate this aspect:

361

362 Preferred solution with respect to Eating SoonTT

363

364 My preferred (fully automatic) solution is to **not** set an EatingSoonTT ahead of the meal (at my normal, even-
365 numbered, profile target). But, as a refinement, I use an Automation:

366 I have my loop **automatically set** a low (even-numbered) TT around the time when it gives first SMBs, i.e. at
367 certain bg delta. Two reasons:

- 368 1. "to orient the calculated insulinRequired towards a more aggressive target"
- 369 2. to get full bgAccel_ISF (rather than, @ glucose < target, half, see "cap_weight" in formula for
370 bgAccel_ISF @ https://github.com/ga-zelle/autolSF/blob/A3.2.0.4_ai3.0.1/autolSF3.0.1_Quick_Guide.pdf,
371 quoted also in beginning of [section 4.2.4](#))

372 The **Automation** must look like: CONDITIONS: likely meal time of day + 1st sign of a beginning meal (e.g.
373 bg delta > 10 mg/dl) + iob under (?) U (to kick in only in initial phase) + no TT running => ACTION: set
374 TT=74 mg/dl for ~30 minutes

375

376 **Discussion ****): This will often **not** influence the first one or two SMBs (which may already come at
377 acceleration detection *before* a + 10 is seen). But it can help "de-couple" treatment of low vs high carb
378 meals:

- 379 • All meals have an acceleration stage in the very beginning, when we already do want SMBs. We should
380 tune bgAccel_ISF_weight (in my suggested mode, without setting a TT) so that all, also low carb meals,
381 get a proper iob boost asap.
- 382 • Only hi carb meals will quickly progress into a +10 mg/dl per 5 minutes rise. The TT kicking automatically
383 in, then, produces two **highly relevant benefits**:
- 384 ○ the next SMBs (driven still by bgAccel_ISF, or already by pp_ISF) get extra boost . . .
385 (see points 1. / 2. discussed above, under "Refinement:"), ... and
- 386 ○ the "dynamic iobTH" is automatically elevated, exactly in the time window where it counts. This
387 means, SMBs are shut off later, and on average you get a bit more iob for high carb meals.
388 ((Sorry, this is a bit pre-mature to discuss here. More see [section 6.1.3](#))).

389 In conclusion, this Automation can bring a gradual improvement regarding the size of e.g. the third and the
390 fourth SMB given after meal start.

391 But this Automation is not essential to have. You could just not worry about EatingSoonTTs and related
392 Automations, and leave eventual implementation of this "refinement" to a later round of fine-tuning.

- 393 • Benefits should also be weighed against **potential draw backs** =
- 394 ○ Any random bg jump will get you aggressive SMBs, whenever the Conditions you defined are
395 given (and there is any insulin required). To prevent that, carefully "tune" jump size, or set a time
396 window etc for this Automation!
- 397 ○ No other Automations that you might have on board, and which require "no TT set" as a condition,
398 will be able to run. (It seems unlikely that you really would like *another* Automation kick in, in that
399 important post-meal phase. But watch what may happen 24/7, considering all your Automations!)

400 ******) Remark: This is a bit pre-mature here. You may need to come back after having worked through [section 4](#).

2.6 Other settings in AAPS/Preferences for autoISF FCL

Make sure you start your migration to FCL with a **solid profile** that worked fairly OK also without a bunch of tricky Automations, and without dynamicISF (which, both, unfortunately, too often are employed to counter-act principle problems with profiles) (or even with technical loop functions, like leaking pods, see [section 1](#)).

Indicators for a solid profile in your HCL:

- Not too short DIA for your fast insulin
- Basal keeps your bg steady in open loop testing
- Meal Management in Hybrid Closed Loop is satisfyingly established, and can serve as a blueprint for your FCL set-up.
- Most important: **ISFs** experimentally proven in relevant times of day, and “working fine” also in Hybrid Closed Loop, and allowed **SMB sizes** opened **up to 120 min** basal.
 - Note that by generally going without any carb inputs, we “give up” the **opportunity** we may have utilized in Hybrid Closed Loop **to differentiate** allowed SMB sizes “after meals” (i.e. with, then, cob>0) vs. maybe a milder reaction in other times using a smaller selected setting for UAM minutes of basal/SMB.

With autoISF, we just pro forma use the max setting, and tune further using the toolbox of SMB_range_extention and the various ...ISF_weights, while observing our glucose curves ([section 4.1.2](#)).

[Section 5](#) will elaborate on how, in FCL, you can differentiate “aggressiveness” even further, if needed in special situations.

One frequently observed “burden” loopers bring with them is that they could *not* operate with 120 minute SMB settings in AAPS Master HCL because they did their ISF tuning wrong. Their lower settings on allowed SMB sizes covered up their principal problem, which now is bound to come up and hurt. See also beginning of [section 4](#).

The following is *not* a list of *all* settings in AAPS / Preferences. We just like to bring up some settings that may not be fully understood, or might interfere “behind your back” with what you try to do.

1. In AAPS Preferences, enable: SMB, SMB with high TT, and SMB always. Also enable UAM, of course. Then go all the way down in all sub-pages of: Open APS SMB / autoISF settings/SMB delivery settings: Enable alternative activation of SMB depending on active target: **ON**

This option, to switch SMB off at any odd-numbered glucose target, makes the restriction, to shut out SMBs at elevated target, unnecessary. It is the reason why “Enable SMB with high TT” is set to **ON**.

We point to this first, because setting an **odd glucose target** in the top right TT field of your AAPS home screen will be a super convenient “**emergency brake**” for you, in your tuning process, to shield yourself against a FCL “going wild” with more SMBs.

If you implemented your iobTH ([section 2.4](#)) well, that iobTH feature serves as a principle, automated, first line of defense against hyper-aggressive SMB fire from your FCL.

The usefulness of the additional, odd target “emergency brake” will, in the long run, lie more in preventing FCL over-reaction to bg bumps that are unrelated to a major meal ([section 5.1](#)).

2. Use Autosens: Should be **OFF**, see the pop-up warning that the feature does not make good sense when entering no carbs. See also remark at point 5.

If for curiosity you want Autosens ON, do so only temporarily and with very narrow Autosens min and max settings, like 0.9 – 1.1. The AAPS main screen shows you the different “opinions” about how ISF now should be modulated by Autosens or by autoISF, and the AUTO ISF tab would show you how both would interact if both active.

Also, do not use Autotune. Enable SMB **always**. (I think for iAPS users we need add: .. and switch off dynamic ISF, dynamic CR, and sigmoid). You may need a look into your CGM whether or how it allows to do SMB always.

3. How frequently SMBs: **3 min**, or for Libre 3 (1 minute): 1 min

4. High TT raises sens: **ON**

This needs to be ON, so via a set half-basal exercise target, loop aggressiveness (profile basal, ISF used) can automatically get significantly **reduced**. Effect is the stronger the lower this parameter is set (in AAPS/Preferences default is 160 mg/dl*), and the higher the TT.

- Details see on Exercise Mode page of the autoISF Quick guide: https://github.com/ga-zelle/autoISF/blob/A3.2.0.4_ai3.0.1/autoISF3.0.1_Quick_Guide.pdf.
- If set *OFF*, an elevated TT has only a “lamer” effect (ISF is then *not* “softened”(raised), loop is just not aiming for the usual profile target with the corrections).

Likewise, you can opt to use this “half basal...” feature to **increase** loop aggressiveness (beyond the basic effect the set low bg target you are shooting for, triggers) in case you switch “low_TT_reduces_sensitivity” ON in /Preferences:

5. Low TT lowers sensitivity: Is **default OFF**. This is to make sure your loop does not go into turbo aggressiveness before you 1) have halfway decent settings tuned in and 2) know more about how your loop operates – or *could* operate “with more bite”. So:

- consult https://github.com/ga-zelle/autoISF/blob/A3.2.0.4_ai3.0.1/autoISF3.0.1_Quick_Guide.pdf.
- ..and **switch to ON when ready!**

6. In any case, leave on **OFF** the other two, that “work the other way around” in that they would make detected sensitivity raise or lower the bg target.

For instance, if you set "resistance lowers target", there is very limited experience with how it interferes with autoISF. (It might for instance result in an unintentional switch to a calculated odd bg target, and disable SMBs).

Pre-autoISF, that setting probably was useful to you when also using Autosens (point 2.).

7. Half basal exercise target: Was already discussed at point 4. Leave it at default **160 mg/dl** as a placeholder- You will set this later in your initial tuning when you get to tune for your favorite kinds of exercise ([section 6.1.3](#)).

Lower numbers (you set for your half-basal ex. target in AAPS/Preferences) have stronger effects

- of reducing basal, and elevating ISF (=sharply will reduce loop aggressiveness), as desirable for exercise (and with your selected TT as a fine-tuning tool).
- and, if desired –see point 5.- also the opposite effects, to increase loop aggressiveness with lowered ISFs at low TTs, as desirable maybe in an "EatingSoonTT" context.

8. Activity modifies sensitivity: **OFF** until you get to [section 6.6](#)

- After giving up Autosens, you might like this feature on board asap. So, you could actually start on section 6.6, and have this setting on **ON**, already before going into section 4.
- Be aware, though, that any set temporary glucose target will interrupt your Activity Monitor. It is generally a good idea to operate with as few Automations as possible when you enter your autoISF tuning phase. (Tipp: Just try to avoid stuff you needed Automations for, while you get your basics right).

9. Advanced settings/Always short avg delta: **OFF** (- unless you need it ON because of jittery CGM. Yes, smoothing can reduce problems, but at the cost of losing time for recognizing true bg movements, as well). Same related to your smoothing selection in AAPS **Configuration builder** / Smoothing: **No smoothing** is the preferred solution there if/when/as long as you have an excellent CGM. Next preferred would be Average smoothing. If single values tend to hop around and cause too big SMBs in your case, you may need to try Exponential smoothing, which gives the nicest bg curves but "iron out" the early indication of a rise, which is so important in a no-user-bolus FCL.

Note re. G7: In contrast to G6, the G7 transmitters do not give smooth(ened) values. Hence you probably should use exponential smoothing in AAPS. (Stay connected with other G7 users in FCL to find out more).

Note re. Libre3 (1 minute): 1 minute CGMs require sophisticated smoothing; as of Dec.2024, autoISF dev is still optimizing the math so smoothing yields results while minimizing the general "time lost" penalty (as is noticeably hurtful when smoothing a 5-minute system).

The author has no experience at all with this sensor. Please stay connected e.g. via <https://discord.gg/tamvvh57Xs>

10. Back to /Preferences / Open APS SMB / Advanced setting: For both safety multipliers go **higher**, probably double, the setting as in the dialogue box recommended (for AAPS HCL), so your FCL loop will be able to do up to 500% TBR in the future.
11. autoISF settings: **Do not activate or change settings until you start [section 4](#)**.
12. Percentage of maxIOB above which SMBs disabled: Put in the number determined in [section 2.4](#) for your iobTH%. or **50** as a placeholder before you get to that section.
13. Protect settings in AAPS/Preferences with a (short) **password**, just to secure that scrolling through the many settings will never make un-intended changes.
14. Eliminate the buttons at the bottom of the AAPS home screen as soon as you can (in AAPS Preferences/Overview/Buttons: all OFF).
- This can strongly enhance personal safety. (Adult: if temp. losing control of phone; kid: if “playing around” on the phone screen).
- Note: The insulin button would allow to start a dangerous insulin shot.
- PS: Pressing on one of the top three buttons would be much less critical: It might require more inputs (like also a duration) to do anything at all. And it would only tweak insulin sensitivity in whatever the loop will do (if anything) in upcoming 5-minute segments.
15. Analyze, in your data, problems you sometimes encounter, or nearly encountered. Then **define alarms** that make your system safer and better. Sophisticated options for alarms are offered by xDrip, and notably when thinking of Automations (with an alarm as Action) in AAPS.

Next steps

To define a reasonable figure for iobTH% and to make a couple of settings for the SMB delivery settings were your preparatory tuning tasks in this [section 2](#).

We suggest to review the basic description of autoISF by ga-zelle

- https://github.com/ga-zelle/autoISF/blob/A3.2.0.4_ai3.0.1/autoISF3.0.1_Quick_Guide.pdf
- and get oriented, directly there in Github, or via [section 3](#), regarding other support materials available from developers

before activating your autoISF for FCL meal management ([section 4](#)).